SC3102: Signals, Systems, and Transforms

Tutorial 01: Discrete signals and systems

1. Consider the following discrete signals:

$$x_1[n] = 4\sin\left(\frac{\pi n}{12}\right), \qquad x_2[n] = \cos(3.5\pi n), \qquad x_3[n] = \cos(n), \qquad x_4[n] = \sin(\pi n) - \cos(n)$$

(a) Determine the fundamental periods of the four signals or explain why they are aperiodic

(b) Compare x_2 to its continuous counterpart $x_{2,c} = \cos(3.5\pi t)$ and explain the discrepancy

(c) What is the period of $x_1 + x_2$? What about $x_3 + x_4$?

(d) Can a non-zero periodic signal be absolutely summable?

2. A sensor is known to provide continuous sinusoidal signals with a waveform matching $x_c(t) = \cos(2\pi f t)$ for some unknown f. We then sample the sensor at a sampling frequency $f_s = 2000$Hz.

(a) What is the sampling interval, T_s?

(b) What is the discrete sequence obtained from sampling the continuous-time signal?

(c) After sampling, we observe the signal $x = \cos(0.1\pi n)$. Does this uniquely determine f?

(d) Suppose additionally, we know $0 < f < 1000$. Can we determine the continuous signal?

(e) Suppose instead $0 < f < 2000$. Is the continuous signal uniquely determined? If not, describe another signal with the same measurements.

3. Suppose a discrete-time system, T, is defined by:

$$(Tx)[n] = x[n] + x[n-1] + x[n]x[n-1].$$

(a) What is the output of the system when the input is the $\delta[n]$?

(b) Is the system causal, stable, and time-invariant? Justify your answer with an explanation. For causality, assume the following definition. For any $N \in \mathbb{Z}$ and two signals x and z:

$$\forall n \le N, x[n] = z[n] \implies \forall n \le N, T(x)[n] = T(z)[n].$$

Informally, if two signals match up to a time point N, then their outputs when fed into the system also match up to N.

(c) Determine if the system is LTI. Provide a counterexample if the system is not LTI.
Hint: Consider input examples with support size 2 *and analyze linearity.*

4. Suppose two LTI systems, T_1 and T_2, have impulse responses:

$$h_1[n] = \delta[n] \text{ and } h_2[n] = u[n]$$

where $\delta[n]$ and $u[n]$ are the impulses and the step functions respectively.

(a) Are the systems causal and stable? Provide a justification.

(b) Using the convolutional formula, determine $T_1 x$ and $T_2 x$. What is $T_2 u$?

(c) Suppose a third LTI system, T, satisfies:

$$(Tu)[n] = 3\delta[n-2] + 3\delta[n-k].$$

What is the impulse response of T?

(d) Is T causal and stable? Does your answer depend on k?